

AI PRODUCT INTELLIGENCE SYSTEM

Day 2 Homework Report

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What I built

- Smart complementary-product recommendations
- Duplicate grouping and a cleaned unique catalog
- Reverse product search from a natural-language query

Real Kaggle run evidence

BACKEND	CATALOG	EMBEDDINGS	MODEL
kaggle-clip	1,331 items	1331, 512	CLIP ViT-B/32

The notebook was executed on Kaggle with the real fashion product images. The report below uses the saved notebook outputs, not sample or placeholder values.

Main result

- Task 1 returned three products with scores from 0.696 to 0.707.
- Task 2 found 109 duplicate groups and reduced the catalog from 1,331 to 1,132 items.
- Task 3 returned five ranked products for the query "blue casual shirt" with scores from 0.257 to 0.269.

Important: the dataset has no purchase history, duplicate labels, or query-relevance labels. I therefore report the real output and all measurable catalog/ranking statistics, and clearly mark precision, recall, F1, Recall@K and full MRR as not computable instead of inventing values.

Approach and shared pipeline

The same image representation supports all three tasks

Dataset and model

- Input: styles.csv and product images from the Kaggle fashion dataset. A balanced sample produced 1,331 valid items.
- Model: openai/clip-vit-base-patch32. Every product image became a normalized 512-number embedding.
- Similarity: cosine similarity, implemented as a dot product after normalization.
- No user interface was built because the assignment asks for the working product-intelligence logic and evidence.

Pipeline

1

Load and validate

Read metadata, locate matching image files, skip broken images.

2

Represent images

Create one normalized CLIP vector per product.

3

Run task logic

Recommendation, duplicate grouping, or text-to-image ranking.

4

Apply final rule

Keep category winners, catalog representatives, or top-ranked search results.

5

Save evidence

Tables, images, CSV outputs and embeddings are produced by the notebook.

Practical safeguards

- Use a minimum score for search so very weak results can be rejected.
- Require matching product category before two images can be called duplicates.
- Use the highest-resolution image as the catalog representative; use the lowest ID only as a repeatable tie-break.
- Review uncommon categories and top similarity pairs manually before using the output in a live catalog.

Task 1 - Smart Product Recommendation

Real product images and CLIP ranking from the executed notebook

Input product: Lino Perros Men Formal Purple Accessory Gift Set

Method: choose complementary target categories, filter valid same-gender or unisex products, calculate 70% text-image relevance plus 30% visual-style similarity, then keep the highest-scoring product from each category.

Complementary recommendations for Lino Perros Men Formal Purple Accessory Gift Set

Revv Men Steel Bangle
score=0.707



Mark Taylor Men Grey Striped Shirt
score=0.706

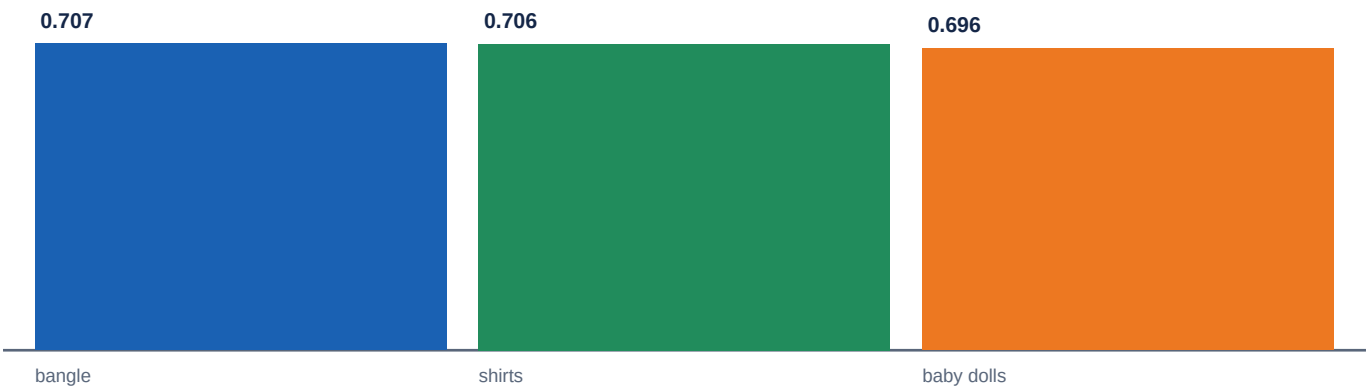


Red Rose Purple Nightdress
score=0.696



Selected product	Category	Final score
Revv Men Steel Bangle	bangle	0.707
Mark Taylor Men Grey Striped Shirt	shirts	0.706
Red Rose Purple Nightdress	baby dolls	0.696

Final recommendation scores



Task 1 - Selection logic and review

Why each product was selected, what worked, and what needs improvement

Final selection logic

- Known product type: use the hand-written complementary-category map.
- Unknown product type: use three frequent categories different from the input category.
- Within each target category: apply the gender filter when possible.
- Final score = $0.70 \times \text{text-to-image relevance} + 0.30 \times \text{visual similarity}$.
- Return one winner per category to avoid three near-identical suggestions.

Measured result

DISTINCT CATEGORY COVERAGE

3 / 3 (100%)

TOP SCORE

0.707

SCORE SPREAD

0.011

Observation

The actual input was an accessory gift set, which is not present in the hand-written complement map. The fallback therefore selected a bangle, a shirt and a baby-dolls/nightdress product. The scores are close together, so the model did not show one very strong winner.

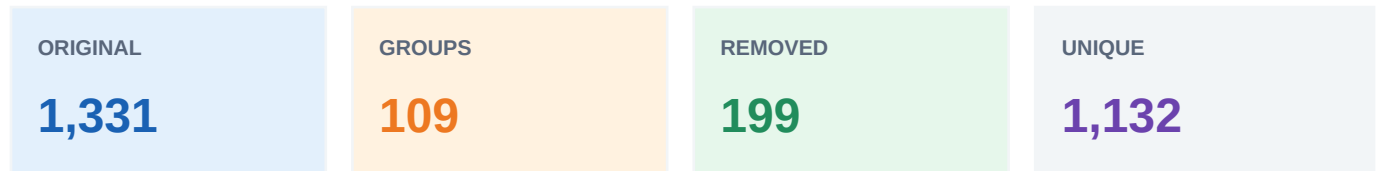
Limitations and practical improvement

- The fashion dataset has no bought-together labels, so recommendation precision or recall cannot be calculated honestly.
- Add rules for accessory gift sets and other uncommon types instead of relying only on frequent-category fallback.
- Use store purchase and click data to learn complementary pairs.
- Show a Needs Review result when scores are close or the input category has no rule.
- Evaluate with category coverage@3 plus human relevance labels, and later report Precision@3, Recall@3 and NDCG@3 from real shopper data.

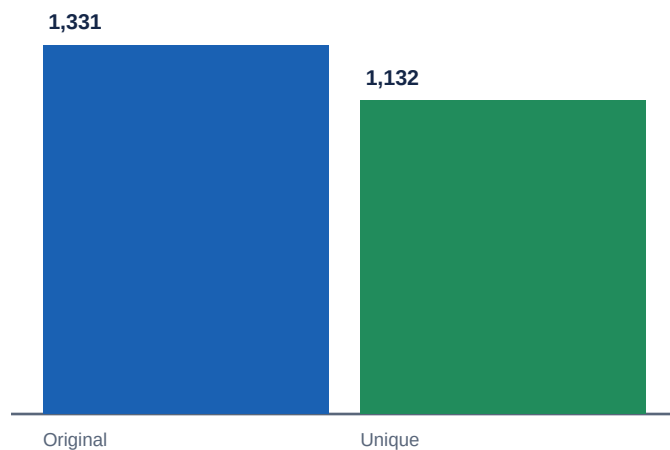
Task 2 - Unique Product Catalog

Duplicate groups, representative rule, and before/after catalog size

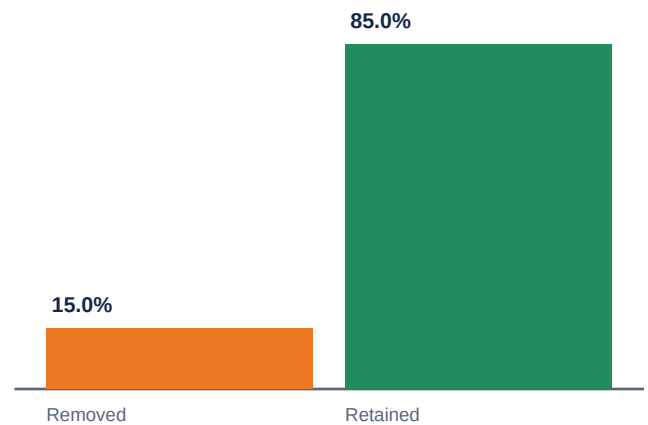
Method: compare image embeddings, accept a duplicate pair only when cosine similarity is at least 0.95 and the product category also matches, join connected pairs into groups, and keep one representative.



Catalog size before and after



Catalog percentage



Representative selection

- For every connected duplicate group, calculate image area = width x height.
- Keep the item with the largest image area because it is likely to be the clearest catalog image.
- If image areas are equal, keep the lowest product ID so repeated runs return the same answer.

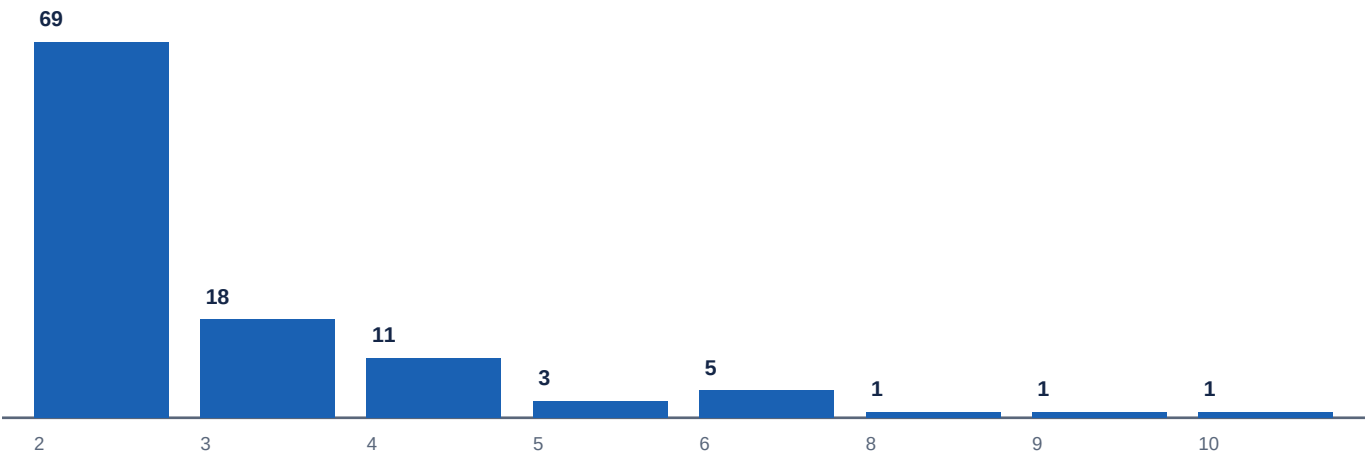
The real run removed 199 entries, a 14.95% catalog reduction, while retaining 85.05% of products.

Task 2 - Duplicate group evidence

Examples from 109 real groups and the complete group-size pattern

Group	Member IDs	Kept ID	Size
342	9041, 4284, 30260, 9702, 17822, 22126, 45587, 11873, 44429, 50419	4284	10
721	47063, 47080, 47065, 47076, 47073, 47071, 47077, 47066, 47068	47063	9
923	28413, 51716, 38210, 16890, 38222, 16879, 28406, 16888	16879	8
91	51390, 51388, 47270, 51391, 53555, 47276	47270	6
768	37528, 37526, 37524, 37525, 37527, 37523	37523	6
795	44560, 35664, 5611, 7330, 33662, 16177	5611	6
935	25763, 25764, 25779, 25765, 25768, 25769	25763	6
974	58905, 58906, 49701, 50281, 40245, 36368	36368	6

Number of duplicate groups by group size



ITEMS INSIDE GROUPS

308

AVERAGE GROUP SIZE

2.83

LARGEST GROUP

10 items

Quality metrics

Metric	Value	Reason / evidence
Catalog reduction	14.95%	Measured from the real run
Retention	85.05%	Measured from the real run
Pairwise precision	N/A	No human duplicate labels
Pairwise recall	N/A	No human duplicate labels
Pairwise F1	N/A	Requires precision and recall

Task 2 - Review, risks and improvements

How I would check the threshold before applying deletion to a live catalog

What the result tells us

- The threshold created 109 groups containing 308 products.
- The catalog fell by 199 products, so every removed row has a retained representative.
- The largest group has 10 items. Large groups need extra review because connected-pair grouping can chain similar but not identical products.

Why precision, recall and F1 are not shown as numbers

Those metrics need a checked list saying which product pairs are truly duplicates. The Kaggle metadata does not include that label. Calling every high-similarity pair correct would make the evaluation circular. I therefore mark these metrics N/A rather than report an unsupported perfect score.

Practical validation plan

- Randomly review at least 100 predicted pairs, including pairs just above 0.95 and pairs from the largest groups.
- Also review likely false negatives just below the threshold, for example 0.90 to 0.95.
- Create human labels: duplicate or not duplicate. Then calculate TP, FP and FN.
- $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$; $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$; $\text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$.
- Tune the threshold by category because shoes, watches and clothing may need different values.
- Never delete automatically on the first run; export a review list and keep an audit trail of the retained ID.

Task 3 - Reverse Product Search

Real text-to-product ranking from the executed notebook

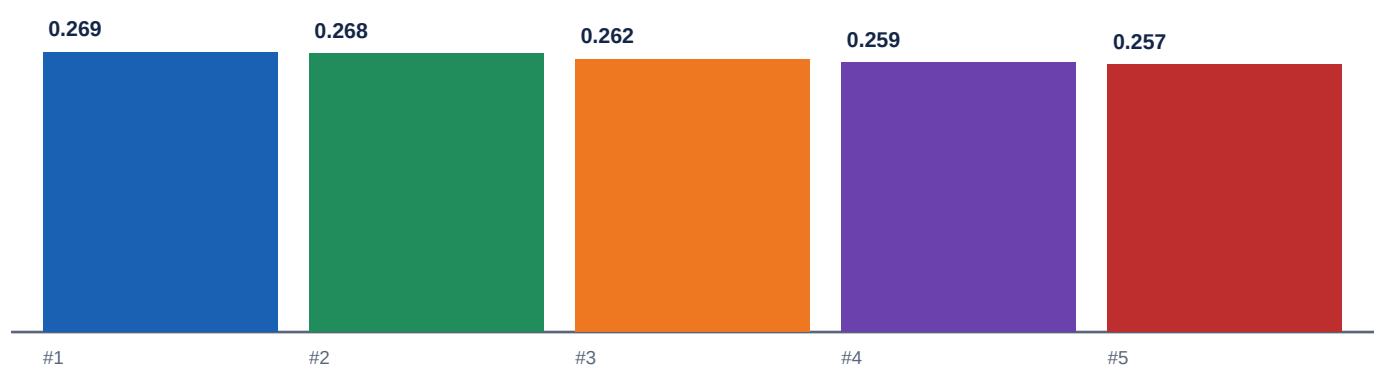
Query: "blue casual shirt"

Method: encode the query with the CLIP text encoder, compare it with all product image embeddings, sort by cosine similarity, reject scores below 0.15, and return the top five.



Ranked product	Type	Color	Score
French Connection Women Blue Top	Tops	Blue	0.269
Forever New Women Vintage Blue Jackets	Jackets	Blue	0.268
Pepe Jeans Men Dark Blue Jeans	Jeans	Blue	0.262
Basics Men White Slim Fit Checked Shirt	Shirts	White	0.259
Lee Men Check Yellow Shirts	Shirts	Yellow	0.257

Text-image ranking scores



Task 3 - Ranking evaluation

Observed quality, Recall@K / MRR requirements, and honest limitations

BLUE MATCH @ 3

3/3 (100%)

SHIRT-TYPE @ 5

2/5 (40%)

STRICT BLUE+SHIRT @ 5

0/5

Observed ranking quality

The first three results match the requested blue color, while two of the five results are shirts. However, no displayed result satisfies both strict conditions (blue and shirt) at the same time. The first result is a blue top, which is visually close but not an exact shirt match.

Metric	Real value	Explanation
Recall@3	N/A	Total relevant catalog IDs were not labeled
Recall@5	N/A	Total relevant catalog IDs were not labeled
Recall@10	N/A	Notebook displayed only top 5 and has no qrels
Full MRR	N/A	Needs a labeled relevant set for each query
Observed MRR@5	0.000	No strict blue+shirt match in displayed top 5
Strict Precision@5	0.000	0 strict matches among 5 displayed results

How I would calculate the requested retrieval metrics

- Prepare 20-50 test queries and ask a reviewer to mark every relevant product ID for each query.
- Recall@K = relevant products found in the top K / total known relevant products. Report Recall@3, Recall@5 and Recall@10.
- MRR = average of $1 / \text{rank}$ of the first relevant result across all test queries.
- Also report Precision@5 and results by query type: color + product, style, gender, and use case.

Final observations and next steps

What worked, current limits, and the most practical improvements

What worked

- The notebook completed on the real Kaggle dataset with CLIP and produced 1,331 image embeddings.
- All three tasks use a clear final-selection rule instead of returning an unexplained model output.
- Recommendation and search outputs include real product images and scores.
- Duplicate cleaning produced a repeatable representative ID and a measurable 14.95% catalog reduction.

Current limitations

- The 1,331-item balanced sample is useful for a first run but not the full dataset.
- Recommendation quality is limited by hand-written complement rules and no purchase-history data.
- Duplicate precision/recall/F1 cannot be measured without checked duplicate labels.
- Recall@3/5/10 and full MRR cannot be measured without query-relevance labels.
- The duplicate threshold is global; some categories may need stricter or looser thresholds.
- The reverse-search example understands color and broad appearance, but the exact blue-shirt intent is not satisfied in the shown top five.

Practical improvements in priority order

1. Add a small human-labeled evaluation set for duplicate pairs and search queries.
2. Tune duplicate thresholds by category and manually review large groups.
3. Add exact metadata filtering or reranking for requested color, product type and gender.
4. Expand complement rules, then replace them with real bought-together data when available.
5. Run the full catalog in batches, save embeddings once, and reuse them for all three tasks.
6. Add a confidence-based Needs Review result instead of forcing a weak recommendation.

Conclusion

The real run proves that the complete pipeline works. The next important step is not a new interface; it is a small, checked evaluation set so the requested precision, recall, F1, Recall@K and MRR can be reported as trustworthy model-quality numbers.